

Topic

Qualitative Analysis of Organic Compounds (Chart)

Compiled by

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Identification of Unknown Organic Compound

The analysis and identification of unknown organic compounds constitutes a very important aspect of experimental organic chemistry. Often, a common first step in the identification of an unknown substance is to determine what elements are present in the sample. Although it is often possible to establish the structure of a compound on the basis of spectra alone (IR, NMR, etc.), the spectra typically must be supplemented with other information about the compound: physical state and properties (melting point, boiling point, solubility, odour, colour, etc.), elemental analysis, and confirmatory tests for functional groups.

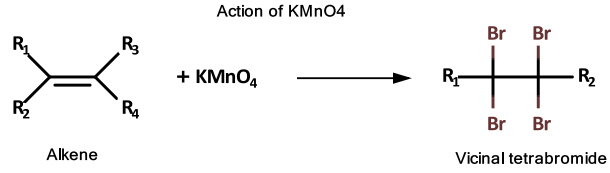
In this experiment you will carry out several qualitative tests that will allow you to identify functional groups in organic molecules. You will then apply what you have learned by characterizing unknown organic compounds in terms of their functional group and solubility behavior.

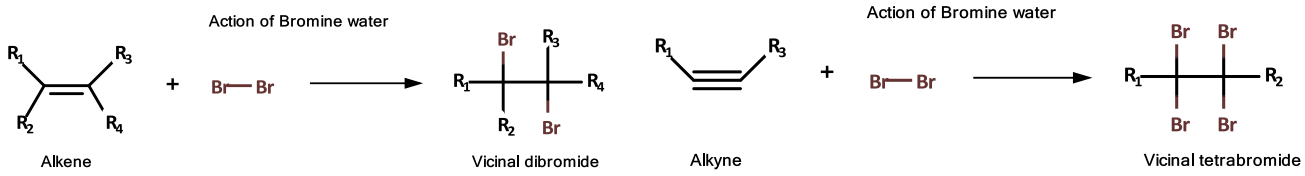
There is no definite set procedure that can be generally applied to organic qualitative analysis. Various books have different approaches, but a systematic approach based on the scheme given below will give good results. Each functional group has a particular set of chemical properties that allow it to be identified. Some of these properties can be demonstrated by observing solubility behavior, while others can be seen in chemical reactions that are accompanied by color changes, precipitate formation, or other visible effects.

In carrying out identification of an organic compound following tests and observations are carried out:

I. Preliminary Tests and Physical Examination
II. Determination of Physical Constants (M.P/B.P)
III. Detection of Elements
IV. Determination of Solubility Group
V. Detection of Functional Group
VI. Special Tests, if any.

	Test	Observation	Inference
I.	**Preliminary Test and Physical Examination**		
A	**State**	i) Solid	Generally high molecular weight, usually having more than 6 carbon atom chain. Eg: Acids, Sugars, Amides, etc.
		ii) Liquid	Generally low molecular weight,. Eg: Alcohols, Ethers, Esters, Aliphatic amines, Aldehydes, Ketones, Hydrocarbons, etc.
B	**Colour**	Pale Yellow	Nitro Compounds s.a. nitrobenzene, dinitrobenzene, nitrophenol, Quinones, Iodoform
		Yellow Orange	Nitro-aniline
		Pink	α-Naphthol, β-Naphthol, resorcinol,
		Red	Azo compounds, methyl orange
		Green	Nitroso compounds
		Greenish Yellow	o/m-nitrobenzaldehyde
		Brown-dark	Phenols, Amines, (darken due to oxidation)
		Colourless	Compounds not containing strong Chromophore. Eg: Acids, Hydrocarbons, ketones, esters, urea, thiourea, acetamide, acetanilide, benzamide, naphalene, etc
C	**Odour**	Pleasant	Alcohols (low mol. Wt.), Aromatic hydrocarbons, ethers, alkyl halides s.a. chloroform, chlorobenzene.
		Deep Sweet	Chloroform
		Fruity	Esters
		Phenolic / Carbolic	Phenols and cresols
		Fishy	Aromatic amines
		Kerosene	Hydrocarbons
		Moth Ball	Naphthalene
		Bitter Almond	Benzaldehyde, nitrobenzene
		Pungent/Irritating	Aliphatic acids, Acid chlorides, formaldehyde, and side chain halo hydrocarbons
	CAUTION: • ***Do not taste an unknown compound.*** • ***To note the odour, cautiously smell the cap of the container and do it only once. Never smell the contents of the container directly.***		

2.	**Ignition Test**		
	Flame Test: Take a small quantity of compound and put it on a metallic spatula or in a porcelain dish and ignite it directly on the	Non-sooty Flame	Aliphatic compounds
		Sooty Flame	Aromatic compounds
		Ammonical odour	Urea, thiourea, amides may be present
		Chars and swells without melting	Sulphur containing compounds
		Melts and chars with a smell of burnt sugar	Carbohydrates may be present
		Produces coughing upon charring	Benzoic acid, salicylic acid etc may be present
		Chars without melting	Sulphanilic acid, starch, uric acid may be present
	Beilstein's Test: Make one end of a copper wire in the form of a loop and heat it on a burner till flame is no longer coloured. Cool the wire and dip the loop in little of the sample and ignite the loop again in flame.	Burns with green flame	Urea, Chlorides, Bromides may be present
		Burns without green flame	Flourides may be present
3.	**Test for Unsaturation**		
	a. **Action of $KMnO_4$(Baeyers test)**: Sub (solid/Liquid) + Sodium carbonate solution + few drops of 2% $KMnO_4$ solution – Shake vigorously	Decolourization of $KMnO_4$	Unsaturated compounds may be present
		No Decolourization of $KMnO_4$	Saturated compounds may be present
	b. **Action of Bromine Water (for freely or sparingly water soluble compounds):** Sub (solid/Liquid) + Bromine water dropwise – Shake vigorously	Decolourization	Unsaturated compounds may be present
		No Decolourization	Saturated compounds may be present
	c. **Action of Bromine Water (for water insoluble compounds):** Sub (solid/Liquid) + 2ml CCl_4 or $CHCL_3$ solution dropwise with shaking + Bromine in CCl_4 dropwise – Shake vigorously.	Decolourization	Unsaturated compounds may be present
		No Decolourization	Saturated compounds may be present
	 Alkene Vicinal tetrabromide		



4. Lassaigne's Test (Elemental Analysis)

Place about a pea size of freshly cut sodium metal into a sodium fusion tube and heat the tube gently to melt the sodium to a shining globule. Add a small quantity of the sample (solid/liquid) into the fusion tube. Heat the tube carefully at first and then as strongly as possible until the bottom of the tube is glowing red, holding the tube at this heat for about 2 min. Quickly plunge the hot tube in a china dish containing about 8 ml of distilled water and cover the china dish immediately with a wire gauge. The tube crumbles into pieces and the mass comes out and dissolves in water. Boil the contents of the china dish thoroughly, filter the contents and collect the filtrate (which is also called Stock solution) in a test tube. Divide the filtrate into 3-4 portions and test each portion for the elements separately.

Test	Observation	Inference
1. Test for Nitrogen:		
a. Prussian Blue test Stock solution + $FeSO_4$ solution – boil and cool. Add conc. H_2SO_4	Green or Blue colour	Nitrogen present
	No Green or Blue colour	Nitrogen absent
2. Test for Sulphur:		
a. Sodium nitroprusside test Stock solution + Sodium nitroprusside	Pink colour	Sulphur present
	No Pink colour	Sulphur absent
b. Lead acetate Test Stock solution + Acetic Acid + 2 ml Lead acetate (5%)	Black Ppt	Sulphur present
	No Black Ppt	Sulphur absent
3. Test for Halogens:		
a. Silver Nitrate test Stock solution + dil. HNO_3 – heat, boil and reduce to half the volume + a few drops of silver nitrate solution.	White ppt, which freely dissolves in 2 ml of ammonia solution	Chlorine present
	Pale yellow ppt, which is difficult to dissolve in 2 ml of ammonia solution	Bromine present
	Yellow ppt, which is insoluble in 2 ml of ammonia solution	Iodine present
	No Ppt	Halogens absent

Test for Nitrogen:

$$6\,NaCN + FeSO_4 \rightarrow \underset{\text{Sod. Ferrocyanide}}{Na_4[Fe(CN)_6]} + Na_2SO_4$$
Sodium cyanide

$$FeSO_4 \rightarrow 2Fe_2(SO_4)_3$$

$$Na_4[Fe(CN)_6] + 2Fe_2(SO_4)_3 \rightarrow \underset{\text{Ferricferrocyanide (Persian Blue)}}{Fe_4[Fe(CN)_6]} + Na_2SO_{4\,dil.\,H}$$

Test for Sulphur:

$Na_2S + Na_2[Fe(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NOS] + 2NaOH$

Sod. nitroprusside — Sod. Ferric sulphocyanide (purple-violet colour)

$Na_2S + Pb(CH_3COO)_2 \rightarrow PbS + 2CH_3COOH$

Lead sulphide

Test for Halogens:

$AgNO_3 + NaCl \xrightarrow{\text{Dil. } HNO_3} AgCl_{(s)} + NaNO_3$

White ppt

$\downarrow 2NH_4OH$

$[Ag(NH_3)_2]^+ Cl^- + 2H_2O$

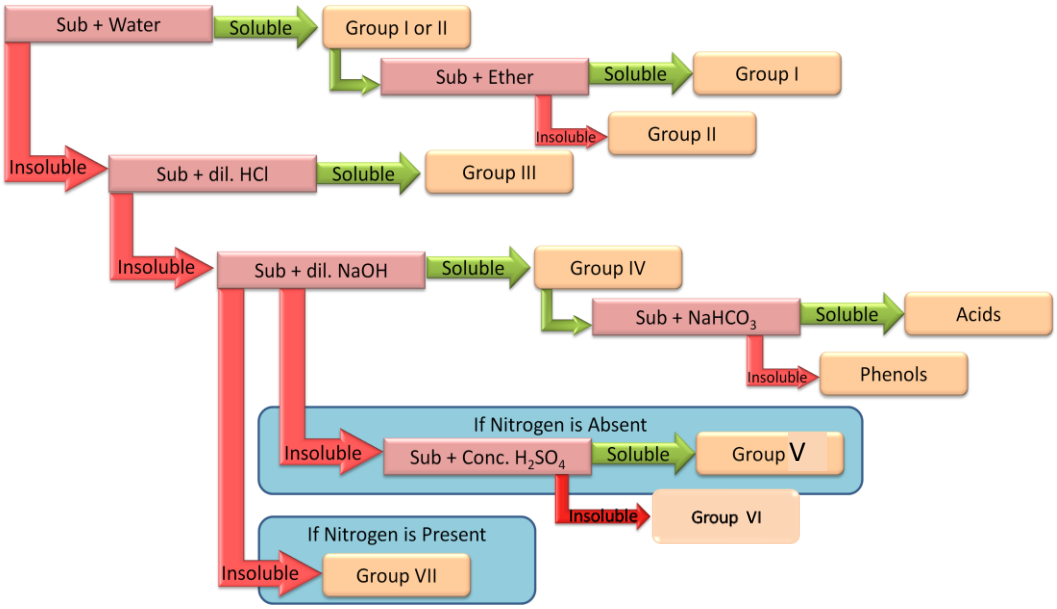
Freely water soluble

5. Solubility test:

a. Sub + Water	Soluble	Group I or II present
Sub + Ether	Soluble	Group I or II present Low molecular weight compounds like acids, esters or resorcinol etc. present
	Insoluble	Group II present Sugars, amides (urea) present
b. Sub + Dil HCl	Soluble	Group III present 1^0, 2^0 or 3^0 Amine present
c. Sub + Dil NaOH	Soluble	Group IV present Acids, Phenols or phenolic acids present
If Nitrogen is Absent		
d. Sub + Conc. H_2SO_4 (In a dry test tube)	Soluble	Group V present
	Insoluble	Group VI present

If Nitrogen is Absent

If all the solubility tests failed and the compound contains Nitrogen, then it belongs to Group VII



Solubility Test:

Substance (Solid/Liquid) + Solvent (Water (ether) : dil. HCl : dil. NaOH ($NaHCO_3$) : Conc. H_2SO_4)

Don't change the above order of solubility test. If the substance is soluble in earlier groups do not proceed further.

<table>
<tr><td colspan="2">Soluble in water</td><td rowspan="2">Soluble in dil. HCl</td><td colspan="2" rowspan="2">Soluble in dil. NaOH</td><td colspan="2">If N is absent</td><td>If N is present</td></tr>
<tr><td colspan="2">In Conc. H_2SO_4
(Use dry test tube)</td><td rowspan="2">Indifferent</td></tr>
<tr><td>Soluble in Ether
(Use dry test tube)</td><td>Insoluble in Ether</td><td>Soluble/Chemical changes like heat or colour development, ppt etc.</td><td>Insoluble</td></tr>
<tr><td>Group I</td><td>Group II</td><td>Group III</td><td colspan="2">Group IV
Sub + Sodium bicarbonate soln</td><td>Group V</td><td>Group VI</td><td>Group VII</td></tr>
<tr><td rowspan="2">Resorcinol
(solid)
(special phenol)

Alcohols
Aldehydes
Ketones and Acids of Low molecular weight</td><td rowspan="2">Sugars

Glucose (solid)

Urea (solid)
(special amide)

Acetamide (solid)

Thiourea</td><td rowspan="2">1° Amines

Aniline (liquid)

Ortho and meta toluidine
(both liquids)

Para toluidine
(solid)

2° Amines
Methyl and Ethyl Aniline
(both liquids)

3° Amines
Dimethyl and diethyl Aniline
(both liquids)</td><td>Soluble</td><td>Insoluble</td><td rowspan="2">Aldehydes

Benzaldehyde
(liquid)

Ketones

Acetophenone
(liquid)

Benzophenone
(solid)

Alcohols

Benzylalcohol
(liquid)

Esters

Methyl and ethyl benzoate
Methylsalicylate
(all liquids)</td><td rowspan="2">Hydrocarbons

Toluene (liquid)
Naphthalene (solid)
Diphenyl(solid)

Halogen hydrocarbons

Chlorobenzene
Bromobenzene
Benzylchloride
(all liquid)</td><td rowspan="2">Amides

Benzamide (solid)

Anilides

Acetanilide (solid)
Benzanilide (solid)

Nitro Compounds

Nitrobenzene
(liquid)
Dinitrobenzene
(solid)
Nitrotoluene
(liquid)
Dinitrotoluene
(liquid)</td></tr>
<tr><td>Acids

Salicylic acid
Benzoic acid
(both solids)</td><td>Phenols

o, m & p – Cresols
(all liquids)</td></tr>
</table>

	Test	Observation	Inference
6.	**Group Detection:**		
A	**Test for Group I:** Low molecular weight compounds like Acids, Alcohols, Aldehydes, ketones, esters, phenols, etc		
a.	Test for Aldehydes: Sub + 2,4-dinitrophenol	Red ppt	Aldehydes or Ketones present
b.	Test for Alcohols: Sub + Sodium metal in a dry test tube	Effervescence	Alcohol present
c.	Test for Acids: Sub + $NaHCO_3$ solution	Effervescence	Acids present
d.	Test for Esters: Sub + NaOH solution + 1 drop of phenolphthalein indicator	Pink colour disappears	Esters present
e.	Test for Phenols: Sub + Alcoholic $FeCl_3$ solution	Violet colour, greenish purple colour, bluish purple colour	Phenols present
B	**Test for Group II:** Sugars and Amides Nitrogen absent-Sugars: Glucose- see aldehydes, under group V; Nitrogen present-Urea/Thiourea,		
a.	***Test for Sugars***		
	i. Molisch Test: Sub + Molisch's reagent + 2 ml Conc. H_2SO_4 along sides of the test tube.	Violet ring at the junction of 2 layers	Sugars present
	ii. Fehling's test: Sub + Fehling's reagent A and B – warm on water bath	Brick Red colour	Reducing Sugar present
	iii. Benedict's Test: Sub + Benedict's reagent - warm on water bath	Orange colour	Reducing Sugar present
	iv. Barfoed's Test: Sub + Barfoed's reagent - warm on water bath	Red colour	Monosaccharide present
	v. Rapid Furfural test: Sub + rapid Furfural reagent + Conc. HCl – boil	Violet Colour	Glucose present
b.	***Test for Amides:***		
	i. Burn substance over an open flame	Burns without soot and leaves white residue	Aliphatic compounds. Urea may be present.
	ii. Sub + NaOH solutions (heat in a dry test tube)	Smell of NH_3 / Red litmus held on top of test tube turns blue.	Amides (urea) may be present.

	iii. Nitrous acid test Dissolve substance in dil. HCl and add to solution of $NaNO_2$ – Boil off nitrogen and cool test tube.	Effervescence due to evolution of Nitrogen	Amides present
		Clear solution	Aliphatic amides present
		White ppt	Aromatic amides present
	iv. Biuret test Place 0.2 g of Urea in a dry test tube and heat until it melts and all NH_3 has evolved – and it resolidifies. Dissolve the residue in few ml of water + 1 ml of dil. NaOH solution and finally add $CuSO_4$ solution dropwise.	Pink, Violet or Blue colour	Urea confirmed.
c.	***Test for Thiourea***		
	Sub + Potassium ferricyanide + dil. Acetic acid – warm slightly	Green colour changing to blue	Thiourea present
C	**Test for Group III: 1^0, 2^0 and 3^0 Amines**		
	i. Sub + Acetyl chloride or Benzoyl chloride (in dry test tube)	Vigorous reaction (solid separates)	1^0 and 2^0 Amines
	ii. Diazotization Test: Sub + dil. HCl till soluble – cool in ice water. Add ice cold solution of $NaNO_2$ in water.	a. Clear solution/Brown oil obtained – further add ice cold β-naphthol in NaOH solution - Orange Dye	1^0 Amines
		b. Yellow oil separates	2^0 Amines
		c. Red/Brown oil - further add NaOH solution - green solid separates	3^0 Amines
	iii. Carbylamine or Isocyanide reaction: Sub + $CHCl_3$ + NaOH solution – Boil continuously	Carbylamine smell	1^0 amine (Aliphatic / Aromatic)
		No Carbylamine smell	2^0 Amines
D	**Test for Group IV: Acids and Phenols**		
	Sub + $NaHCO_3$ solution	Compound dissolves with Effervescence of CO_2	Carboxylic Acids may be present
		No Effervescence	Phenols present
a.	***Test for Acids: Salicylic acid, Cinnamic acid, Benzoic acid***		
	Confirmatory test for Acids: (Esterification reaction)		
	Sub + 1 ml of Alcohol + Conc. H_2SO_4 (few drops) – heat for some time on water bath, cool and pour in a china dish containing water	Fruity odour due to ester formation	Acid confirmed
	i. Sub + Alcoholic $FeCl_3$	Purple-Violet colour	Phenolic acids – Salicylic acid present
		No colour, but ppt	Salicylic acid absent. Benzoic acid/Cinnamic acid may be present

	ii. Sub + Bromine water – dropwise with shaking	Decolourisation	Cinnamic acid
		No Decolourisation	Benzoic acid
	iii. Fluorescein Test: 0.2 g Sub + 0.2 g resorcinol – heat the mixture + drops of Conc. H_2SO_4 + 1 ml of water – heat (to dissolve the residue) and add excess of NaOH dropwise.	Dark red colouration with green fluorescence (fluorescence can be observed by addition of reaction mixture in to a beaker with 10ml water)	Phthalic acid/succinic acid may be present
		Dark red colour with green fluorescence not obtained	Benzoic acid is present
b.	***Test for Phenols: Cresols, Resorcinol***		
	i. Sub + Alcoholic $FeCl_3$	Purple/Green/Violet colour	Phenols or Cresols present
	ii. Libermann's Nitroso test: Add little of phenolic compound to minute crystal of $NaNO_2$ (sodium nitrite) in a dry test tube – Heat gently for a minute, cool and add few drops of Conc. H_2SO_4 from the sides	Green/Blue colour with water Red colour with dil. HCl Green colour with dil. NaOH	Phenolic compound confirmed
	iii. Phthalein Test: 0.2 g Sub + 0.2 g Phthalic anhydride + 2-3 drops of Conc. H_2SO_4 (in a dry test tube) – heat a min and cool – add 1 ml of water then pour in dil. NaOH solution	Different colours	Phenolic compound confirmed
		Violet / pink colour	*Phenol*
		Intense violet/red	*o-cresol*
		Bluish purple	*m-cresol*
		Pink-red / light green	*p-cresol*
		Red colour with green fluorescence	*Resorcinol*
		Greenish blue / dark green	*α- naphthol*
		Permangnate colour	*β-naphthol*
		No colour / no reaction	*Salicylic acid*
E	**Test for Group V: Aldehydes, Ketones, Alcohols, Esters**		
a.	***Test for Aldehydes***		
	i. Sub + 2,4-dinitrophenylhydrazine solution (2,4-DNP)(excess) – warm if necessary	Precipitate (red/Orange/Yellow)	Aldehyde/Ketone present
	ii. Tollen's Test: Sub + Tollen's reagent (ammonical $AgNO_3$ soln) – warm on a water bath	Silver mirror formed on the walls of the test tube/ grey-black ppt	Aldehydes present (aromatic aldehyde gives test)
		No Silver mirror	Ketones present
	iii. Sub + Fehling's solution (A+B) – warm on a water bath	Red ppt	Aldehyde present (benzaldehyde does not give red ppt)
	iv. Sub + Schiff's reagent (shake well & observe the colour after 3-4 min)	Violet-purple/Magenta colour (Note: Light pink colour should be considered as negative result)	Aldehyde confirmed (glucose does not give pink colour)
b.	***Test for Ketones***		
	i. Sub + 2,4-dinitrophenylhydrazine solution (excess) – warm if necessary	Precipitate (red/Orange)	Aldehyde/Ketone present

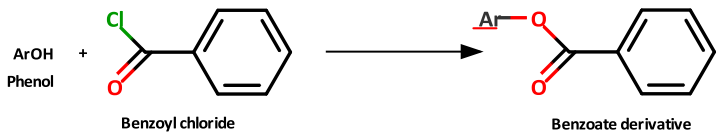
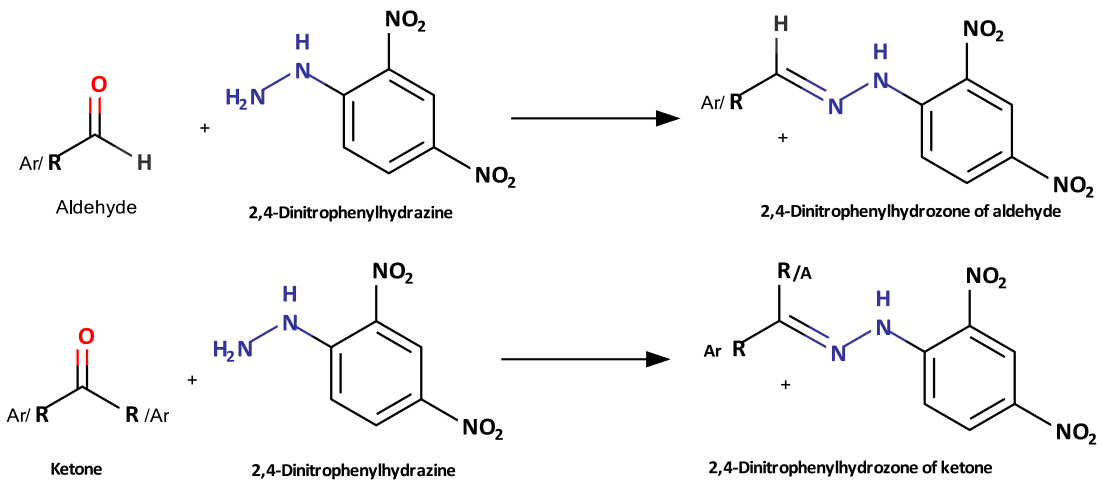
	i. Tollen's Test: Sub + Tollen's reagent (ammonical $AgNO_3$ soln) – warm on a water bath	No Silver mirror	Ketones present
	ii. Sub + Fehling's solution (A+B) – warm on a water bath	No Red ppt	Ketone present
	iii. Sub + Schiff's reagent (shake well)	No Violet-purple/Magenta colour	Ketone present
	iv. Sub + freshly prepared Sodium nitroprusside solution – shake well + dil. NaOH solution in excess	Pink/Red colouration	Ketone with methyl group(-CO-CH_3) (Acetophenone/Acetone)
	v. Sub + 0.2 g naphthalene (in a dry test tube) – heat until molten + small dry piece of Sodium – heat.	Surface of Sodium appears greenish-blue	Benzophenone confirmed
c.	***Test for Alcohols***		
	ii. Sub + 2,4-dinitrophenylhydrazine solution (excess) – warm if necessary	No Red Precipitate	Aldehyde/Ketone absent Alcohol may be present
	iii. Sub + freshly cut shiny Sodium metal (in a dry test tube) - after complete dissolution of sodium add little ether	Vigorous reaction (brisk effervescence of hydrogen gas) - Formation of hazy solution / solid salt after addition of ether	Alcohol present
	iv. Esterification reaction: Sub + Benzoic acid + Conc. H_2SO_4 (in a dry test tube) boil for some time and pour in a china dish containing water or Na_2CO_3 soln.	Fruity Odour	Alcohol confirmed
	v. Sub + Conc. H_2SO_4 (in a dry test tube)	Tube becomes hot and a white gelatinous product is obtained	Benzyl alcohol confirmed.
d.	***Test for Esters***		
	i. Sub + 2,4-dinitrophenylhydrazine solution (excess) – warm if necessary	No Red Precipitate	Aldehyde/Ketone absent Alcohol/Esters may be present
	ii. Sub + Sodium metal (in a dry test tube)	No Reaction	Alcohol absent Esters may be present
	iii. Sub + dil. NaOH Solution + drop of phenolphthalein – pink colour – shake continuously for a few minutes/heat on flame	Pink colour disappears	Ester present
	iv. Hydroxamic acid test: Sub + 1 ml of methanolic hydroxylamine hydrochloride + methanolic KOH/NaOH – boil on a water bath – cool and acidify with dil. HCl + $FeCl_3$ solution	Pink colour	Ester present
F	**Test for Group VI: Hydrocarbons, Halogen hydrocarbons**		
a.	***Test for hydrocarbons***		

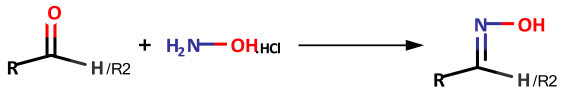
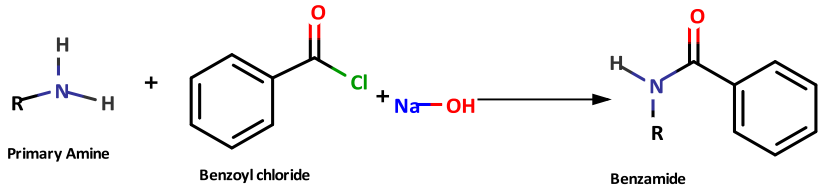
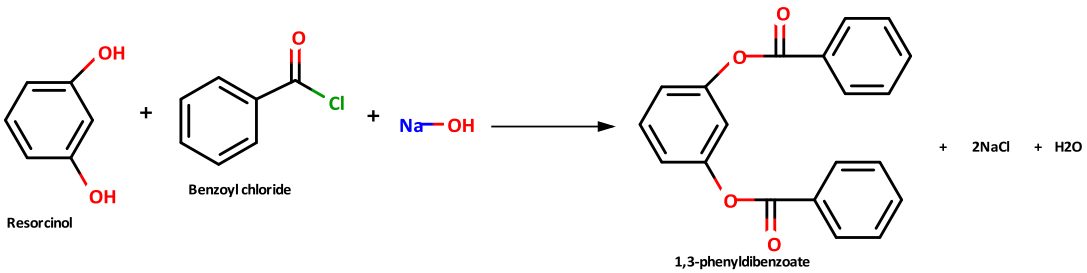
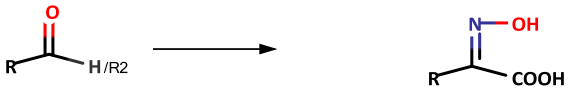
	i. Sub (In a dry test tube) + Conc. H_2SO_4 – heat till a clear solution is obtained – cool and pour in a beaker of water	Clear solution (no oil layer/solid separates)	Hydrocarbons present (sulphonation takes place)
	ii. Formalin test (Le Rosen test for Aromatic compounds) In a porcelain dish take 0.6 g of Sub in $CHCl_3$ + 4-5 drops of formaldehyde-sulphuric acid reagent. [Reagent: 5-6 drops of 37% HCHO + 5 ml of Conc. H_2SO_4. It should be prepared freshly]	Different colours like Red, Green etc.	Toluene, Benzene, Naphthalene present
	iii. Chloroform-Aluminium Test: Heat strongly 0.2 g of $AlCl_3$ in a test tube so that it sublimes and sticks on the walls of the test tube. Run down 2 ml of Sub in $CHCl_3$ on the sides of the test tube – note the colour produced by contact of solution of $AlCl_3$	Appearance of red, orange, blue or green colour due to the formation of triphenylmethane dyes.	Aromatic hydrocarbons confirmed
b.	***Test for halogen hydrocarbons***		
	i. Elemental Analysis (Lassaigne's test)	Halogen present. Chlorine/Bromine	Halogen hydrocarbon
	ii. Sub + alcoholic $AgNO_3$ solution – Warm	White ppt	Halogen present in side chain
		No white ppt	Halogen present in nucleus
	iii. Test for Alkyl/Aryl halides Sub + 2 ml Conc. HNO_3 + 2 ml Conc. H_2SO_4 – warm gently for 3 mins and pour in a beaker of water	Yellow ppt	Indicates arylhalides
G	**Test for Group VII: Amides, Anilides, Nitro compounds**		
a.	***Test for Amides***		
	i. Sub + NaOH solution – heat	Evolution of NH_3 detected by smell or red litmus turned blue	Amide present
		No smell of NH_3	Amides absent
	ii. Nitrous acid test: - Place and dissolve little of compound in dil. HCl and add to solution of $NaNO_2$ - Boil off nitrogen and cool the test tube	Effervescence due to evolution of nitrogen	Amides present
		Clear solution	Aliphatic amide present
		White ppt	Aromatic amine present
b.	***Test for Anilides***		
	i. Sub + NaOH + $CHCl_3$ – boil for some time	Smell of carbylamine	Anilide Present
		No Smell of carbylamines	Anilide absent

	ii. Sub + Conc. HCl – boil for some time till clear solution is obtained. Dilute with water – cool + ice cold $NaNO_2$ solution. Finally add ice cold β-naphthol in NaOH in solution dropwise	Red/Orange dye	Anilide confirmed
c.	***Test for Nitro Compounds***		
	i. Sub + NaOH solution – boil	No smell of NH_3	Amides absent
	ii. Sub + NaOH solution + $CHCl_3$ – boil continuously for some time	No smell of carbylamines	Anilide absent
	iii. Sub + Conc. HCl + Tin pieces – wait till reaction ceases – then boil till layer disappears – cool, filter, dilute and add ice cold $NaNO_2$ solution	Clear solution - Further add ice cold β-naphthol in NaOH in solution dropwise. - Red/Orange dye obtained	Nitro compound present
	iv. Neutral Reduction: Dissolve 0.1 g of sample in 2 ml ethanol + 5 drops of $CaCl_2$ solution + pinch of Zinc dust and boil the contents for 5 mins. Filter the solution in a test tube containing 1 ml of Tollen's reagent	Grey/Black ppt (NO_2 group present)	Nitro compound present
	v. Test for Aromatic nitro compounds: Sub + acetone + NaOH solution	Pink colour	Dinitro compound
		No pink colour	Mononitro compound

Derivative Preparation

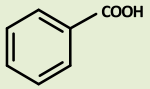
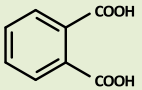
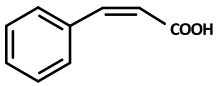
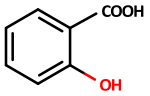
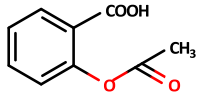
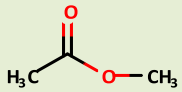
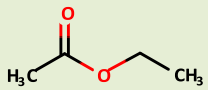
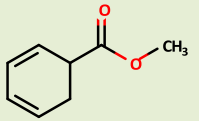
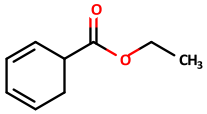
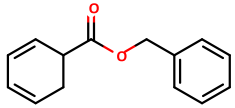
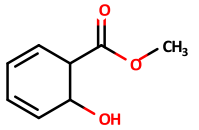
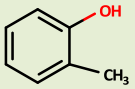
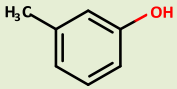
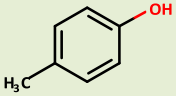
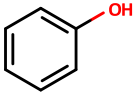
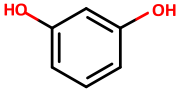
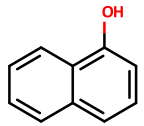
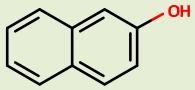
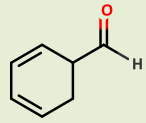
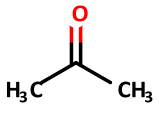
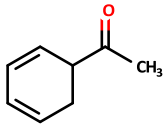
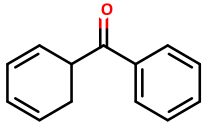
Derivatizations are the synthetic procedures regarding 'conversions' of one simple common organic compound into another compound that can be thoroughly characterized. After performing the preliminary and elemental analysis, solubility tests, functional groups test and determination of melting/boiling point, the student can propose a list of possible compounds for the given unknown sample. These possible compounds may contain a number of structural differences; therefore the final confirmation for the identity of the given unknown sample can be accomplished by preparation of a suitable derivative of the given sample and noting its melting/boiling point.

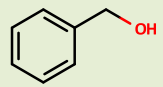
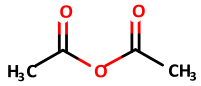
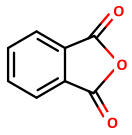
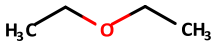
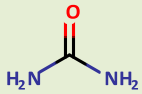
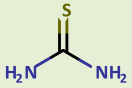
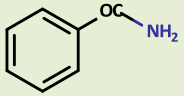
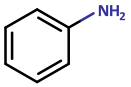
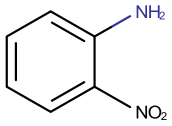
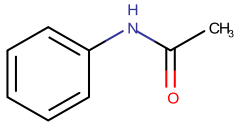
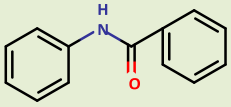
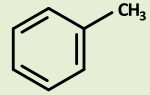
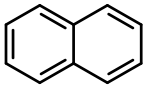
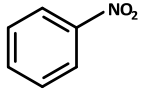
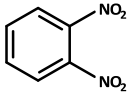
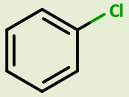
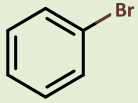
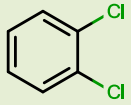
Sr. No.	Procedure
1.	***Urea Nitrate (For Urea)***
	To a saturated solution in water add Conc. Nitric acid – Shake well. Crystals of Urea nitrate separate.
2.	***Benzoate Derivative (For Alcohols/Phenols)***
	Place 0.5 g of sample in 2.5 ml of water in a test tube. To this add 2.5 ml of 10% NaOH, followed by by 0.3 ml of benzoyl chloride. Shake for several mins. (or boil on low heat). The odour of benzoyl chloride should disappear. The mixture is poured into water and the product is recrystallized from alcohol.  Benzoyl chloride Benzoate derivative (Caution: Benzoyl chloride is highly corrosive and irritating – Handle with care)
3.	***2,4-Dinitrophenylhydrazone (For Aldehyde/Ketone)***
	Dissolve 0.1 g of aldehyde/ketone in 5 ml of alcohol. To this add 2 ml of 2,4-DNP reagent. Shake vigorously and allow the resulting mixture to stand for some time at room temperature. Crystallization of 2,4-dinitrophenylhydrazone usually occurs within 5-10 mins. The ppt is recrystallized from alcohol. (*2,4-DNP reagent: 0.4 g 2,4-DNP + 2 ml Conc.H_2SO_4 + 3 ml water – Shake well until 2,4-DNP is dissolved – To this add 5 ml 95% alcohol*)  Aldehyde 2,4-Dinitrophenylhydrazine 2,4-Dinitrophenylhydrozone of aldehyde Ketone 2,4-Dinitrophenylhydrazine 2,4-Dinitrophenylhydrozone of ketone

4.	***Oxime Derivative (Aldehydes/Ketone)***
	Dissolve 0.8 g of hydroxylamine hydrochloride in 5 ml of water. Add 4 ml of 10% NaOH solution and 0.4 g of unknown aldehyde/ketone (If sample is insoluble in water, add sufficient ethanol to give a clear solution). Warm the mixture on a water bath for 10-15 mins and allow the product to stand in an ice bath so that crystalline product is obtained from water or dilute ethanol. 
5.	***Benzoate Derivative – Schotten-Baumann reaction (For 1°/2°/3° Amines, Phenols, Sulphonamide)***
	Dissolve 0.5 g of unknown compounds in 10 ml of 5% NaOH. Add 0.5 g of Benzoyl chloride with continuous shaking for 20 mins – Allow to cool. The reaction mixture is again vigorously shaken for 15 mins – Ppt obtained is filtered off and recrystallized in alcohol.  Primary Amine Benzoyl chloride Benzamide  Resorcinol Benzoyl chloride 1,3-phenyldibenzoate (Caution: Benzoyl chloride is highly corrosive and irritating – Handle with care)
6.	***Base Catalyzed Derivative (For Amides/Anilides/Imides/nitriles)***
	Reflux a mixture of 0.5 g unknown sample and 15 ml of NaOH for 15-20 mins. The aqueous solution is allowed to cool in ice bath. Add conc. HCl drop-wise until the mixture becomes acidic to litmus. Filter the solid carboxylic acid ppt. (*If no ppt obtained extract the mixture thrice with 8 ml diethyl ether and pass the combined extract over anhydrous magnesium sulphate* and evaporate the ether at room temp to leave the acid residue.) 
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Physical Constants of Compounds and their derivatives

Name of the Compound [M.P./B.P. (°C)]	Structure	Derivative with [M.P./B.P. (°C)]	Name of the Compound [M.P./B.P. (°C)]	Structure	Derivative with [M.P./B.P. (°C)]	Name of the Compound [M.P./B.P. (°C)]	Structure	Derivative with [M.P./B.P. (°C)]
Acetic acid B.P – 118		Phenylhydrazine [129] Amide [82]	Benzoic acid M.P - 121		Phenylhydrazine [167] Amide [139] Anilide [162]	Phthalic acid M.P – 208		Amide [219] Anilide [251]
Cinnamic acid M.P - 133		Amide [153] Anilide [147]	Salicylic acid M.P – 158		Amide [135] Anilide [139]	Acetylsalicylic acid M.P – 135		Amide [135] Anilide [139]
Methyl acetate B.P – 57		Amide [82] 3,5-Dinitrobenzoate [108]	Ethyl acetate B.P – 77		Amide [82] 3,5-Dinitrobenzoate [93]	Methyl benzoate B.P – 199		Amide [130] 3,5-Dinitrobenzoate [108]
Ethyl benzoate B.P – 212		Amide [130] 3,5-Dinitrobenzoate [93]	Benzyl benzoate B.P – 323		Amide [130] 3,5-Dinitrobenzoate [113]	Methyl salicylate B.P – 224		Amide [142] 3,5-Dinitrobenzoate [108]
o-cresol M.P – 30 B.P -191		p-Nitrobenzoate [94]	*m*-cresol M.P – 12 B.P -202		p-Nitrobenzoate [90]	*p*-cresol M.P – 36 B.P -202		p-Nitrobenzoate [98]
Phenol M.P- 30 B,P – 182		Benzoate [69]	Resorcinol M.P – 110		Benzoate [117]	α-naphthol M.P – 94		Benzoate [56]
β-naphthol M.P – 123		Benzoate [107]	Formaldehyde B.P -	HCHO	-	Benzaldehyde B.P – 179		2,4-dinitrophenyl hydrazone [237]
Acetone B.P – 56		2,4-dinitrophenyl hydrazone [128]	Acetophenone B.P – 202		2,4-dinitrophenyl hydrazone [250]	Benzophenone M.P – 49 B.P – 306		2,4-dinitrophenyl hydrazone [238]

Methyl alcohol B.P -	CH_3-OH	-	Ethyl alcohol B.P -	C_2H_5-OH	-	Benzyl alcohol B.P – 205		p-3,5-Dinitrobenzoate [113]
Acetic Anhydride B.P – 140		Acetanilide [114]	Phthalic Anhydride M.P – 132		Anilide [170]	Diethyl ether B.P – 34		-
Urea M.P – 132		Nitrate [163]	Thiourea M.P – 182		Hydrochloride [136]	Benzamide M.P – 130		Benzoic acid [122]
Aniline B.P – 183		Acetamide [114]	Nitroaniline M.P – 114		Acetamide [155]	Acetanilide M.P – 114		*p*-bromo [167]
Benzanilide M.P – 160		Benzoic acid [122]	Benzene B.P – 80		1,3 – Dinitro [90]	Toluene B.P – 111		2,4-dinitro [71]
Naphthalene M.P – 80		1-nitro [61]	Nitrobenzene B.P – 211		Acetamide [114] Dinitro [91]	m-dinitrobenzene M.P – 91		2,4-dinitrophenol [114]
Chlorobenzene B.P – 132		2,4-dinitro [52]	Bromobenzene B.P – 156		2,4-dinitro [75]	Dichlorobenzene B.P – 179		2,4-dinitro [110]

Physical Constants

Sr. No.	Solids	Melting Point (M.P) (°C)
1	Resorcinol	110
2	*p*-toluidine	44
3	Glucose	144
4	Glucose hydrated	90
5	Urea	132
6	Salicylic acid	158
7	Cinnamic acid	133
8	Benzoic acid	120
9	Benzophenone	40
10	Naphthalene	80
11	Diphenyl	70
12	*p*-dichlorobenzene	53
13	Benzamide	128
14	Acetanilide	114
15	Benzanilide	160
16	Dinitrobenzene	90
17	Dinitrotoluene	72
18	Thiourea	172
19	*o*-nitrophenol	45
20	Diphenyl amine	54
21	*p*-nitrophenol	114
22	Benzophenone	49
23	*m*-dinitrobenzene	90
24	Phthalic acid	208
25	α-naphthol	94
26	Lactose	223
27	Anthracene	240
28	*p*-nitroaniline	148
29	Sulphanilamide	165
30	Acetlysalicylic acid	135

Sr. No.	Liquids	Boiling Point (B.P) (°C)
1	Aniline	184
2	*o* toluidine	199
3	*m*-toluidine	202
4	Methylaniline	192
5	Ethylaniline	204
6	Dimethylaniline	292
7	Diethylaniline	215
8	Phenol	182
9	*o*-cresol	191
10	*m*-cresol	203
11	*p*-cresol	202
12	Benzaldehyde	179
13	Acetophenone	202
14	Benzyl alcohol	206
15	Methylbenzoate	199
16	Ethylbenzoate	213
17	Toluene	110
18	Chlorobenzene	132
19	Bromobenzene	155
20	*o*-dichlorobenzene	179
21	Benzyl chloride	179
22	Nitrobenzene	210
23	*o*-nitrotoluene	222
24	*n*-butanol	118
25	Bromobenzene	156
26	Benzene	80
27	Xylene	144
28	Ethyl acetate	77
29	Chloroform	61
30	Acetic acid	118